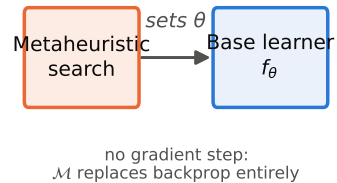
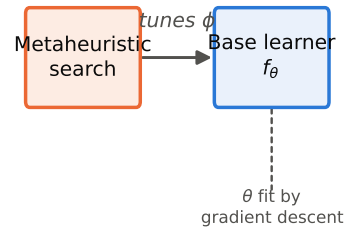


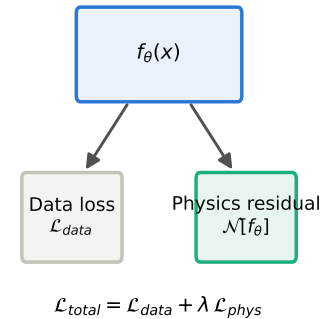
Schematic architecture of the seven hybridisation types (H1-H7)

General couplings each type names (§4.1); not one specific paper's implementation.

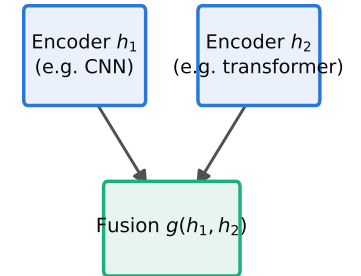
H1 Metaheuristic → hyperparameters H2 Metaheuristic → weights



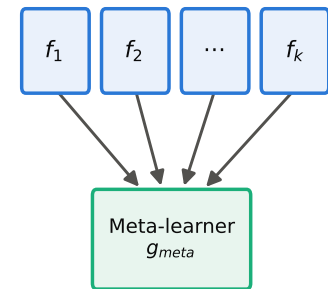
H3 Physics + data



H4 Architecture fusion

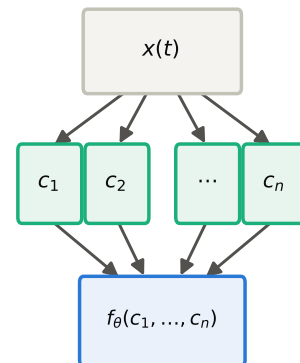


H5 Heterogeneous stacking

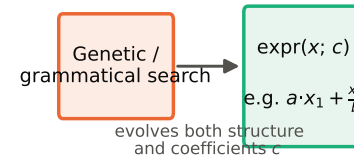


on out-of-fold predictions, not in-fold

H6 Decomposition then learn



H7 Symbolic-numeric



Legend

Base learner / model component

Metaheuristic / search component

Data, physics or decomposition component

Non-learned intermediate quantity